Seeing and hearing

16a

1 Our eyes and ears are sense organs. Name three other sense organs.

_____

2 Sense organs respond to a stimulus. Eyes and ears respond to different waves. The eye responds to _____ and the ear responds to _____.

3 **a** State which type of wave always requires a medium for carrying it.

_____

b State the type of wave that does not always require a transmission medium.

_____

4 Match each ear part with a description and a function, by writing the letters identifying the Description and Function in the Label box. (Add letters to the label cells.)

Labels

1 Anvil __ __
2 Auditory nerve __ __
3 Cochlea __ __
4 Ear canal __ __
5 Ear shell __ __
6 Eardrum __ __
7 Eustachian tube __ __
8 Hammer __ __
9 Stirrup __ __
10 Window __ __

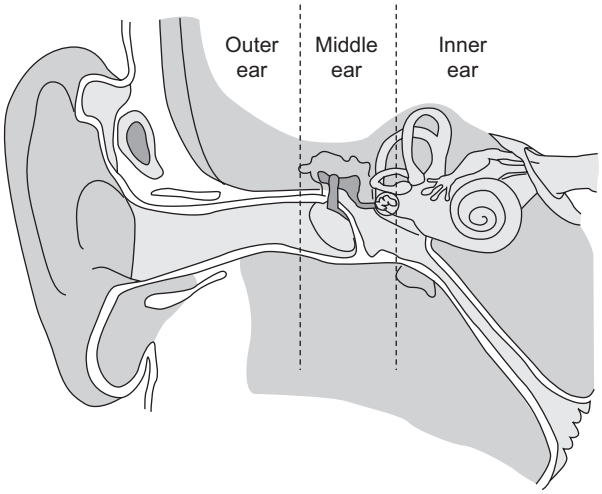
Descriptions

A Shell-like cartilage structure
B Small thin membrane at beginning of cochlea
C Snail-like structure containing a liquid
D Thick nerve linking cochlea and brain
E Thin membrane at end of ear canal
F Thin tube between middle ear and mouth
G Tiny ear bone attached to eardrum
H Tiny ear bone attached to window
I Tiny ear bone between hammer and stirrup
J Tube between ear shell and ear drum

Functions

a Changes vibrations in stirrup to vibrations inside cochlea
b Channels sound waves to eardrum
c Collects sound, directs it to ear canal
d Equalises air pressure between middle ear and mouth
e Holds nerve endings which are sensitive to vibrations
f Magnifies sound vibrations
g Magnifies sound vibrations
h Magnifies sound vibrations
i Sends electrical pulses from nerve endings in cochlea to brain
j Vibrates when sound waves hit it, makes hammer vibrate

5 Add labels to the diagram of the human ear. (All the names can be found in the boxes above.)



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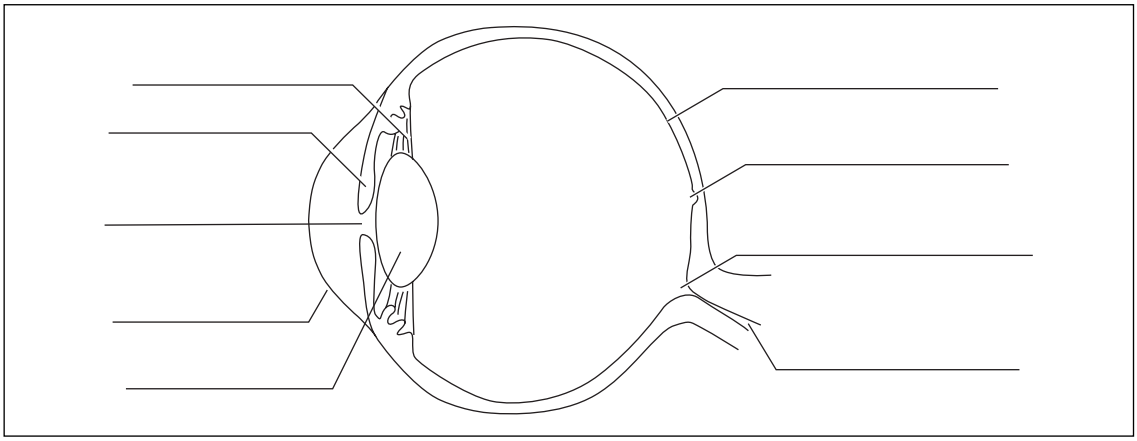
Seeing and hearing

16b

6 Match an eye part with a description and a function. (Add letters to the label cells.)

Labels		Descriptions	Functions
1 Blind spot ___ ___		**A** Area where most light rays hit retina	**a** Adjusts lens for better focusing
2 Cornea ___ ___		**B** Area where optic nerve leaves eye	**b** Allows light to enter eye
3 Fovea ___ ___		**C** Clear skin at front of eye	**c** Carries electrical impulses from retina to brain
4 Iris ___ ___		**D** Clear soft thick disk	**d** Controls amount of light entering eye
5 Lens ___ ___		**E** Colourful broad ring in front of lens	**e** Changes light into electrical impulses
6 Lens muscle ___ ___		**F** Opening controlled by iris	**f** Does not have light sensitive nerve endings, does not 'see'
7 Optic nerve ___ ___		**G** Light sensitive layer inside eye	**g** Has many light sensitive nerve endings and 'sees' well
8 Pupil ___ ___		**H** Ring muscle around lens	**h** Helps focus light rays onto retina
9 Retina ___ ___		**I** Thick nerve linking eye and brain	**i** Protects eye, helps focus light rays onto retina

7 Label the diagram of a human eye. (The names of the parts are in the table above.)



8 State in which way the following are similar:

a Optic nerve and auditory nerve _____

b Retina and cochlea _____

c Lens and hammer, anvil, stirrup. _____

9 Contrast the following:

a Retina and cochlea _____

_____

b Lens and hammer, anvil, stirrup _____

_____

10 Parts in our eye allow us to control things. In your science book, list several parts of the eye you labelled in question 7, and state what they control.

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